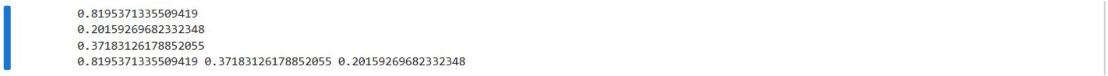
第一题：

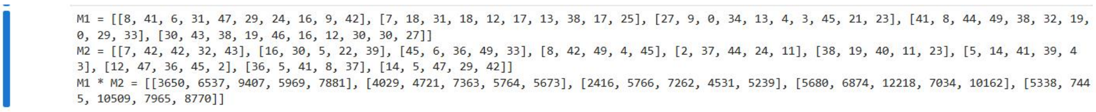
运行结果：



```
0.8195371335509419
0.20159269682332348
0.37183126178852055
0.8195371335509419 0.37183126178852055 0.20159269682332348
```

第二题：

1、运行结果：

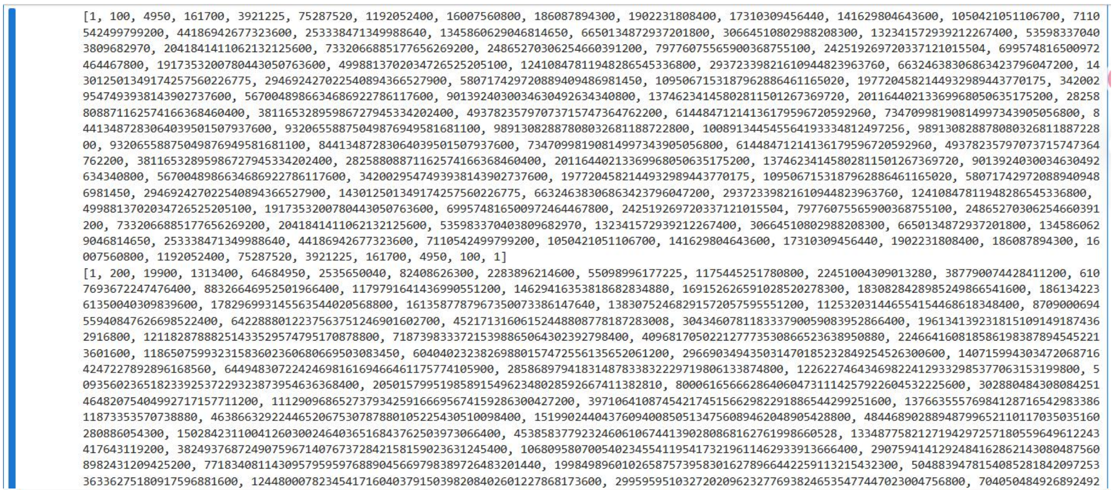


```
M1 = [[8, 41, 6, 31, 47, 29, 24, 16, 9, 42], [7, 18, 31, 18, 12, 17, 13, 38, 17, 25], [27, 9, 0, 34, 13, 4, 3, 45, 21, 23], [41, 8, 44, 49, 38, 32, 19,
0, 29, 33], [30, 43, 38, 19, 46, 16, 12, 30, 30, 27]]
M2 = [[7, 42, 42, 32, 43], [16, 30, 5, 22, 39], [45, 6, 36, 49, 33], [8, 42, 49, 4, 45], [2, 37, 44, 24, 11], [38, 19, 40, 11, 23], [5, 14, 41, 39, 4
3], [12, 47, 36, 45, 2], [36, 5, 41, 8, 37], [14, 5, 47, 29, 42]]
M1 * M2 = [[3650, 6537, 9407, 5969, 7881], [4029, 4721, 7363, 5764, 5673], [2416, 5766, 7262, 4531, 5239], [5680, 6874, 12218, 7034, 10162], [5338, 744
5, 10509, 7965, 8770]]
```

2、我是通过阅读 5. 数据结构 — Python 3.15.0 文档（按住 Ctrl 单击即可访问）中 5.1.4. 嵌套的列表推导式获得灵感，参考如何建立×行×列的矩阵。

第三题：

运行结果：



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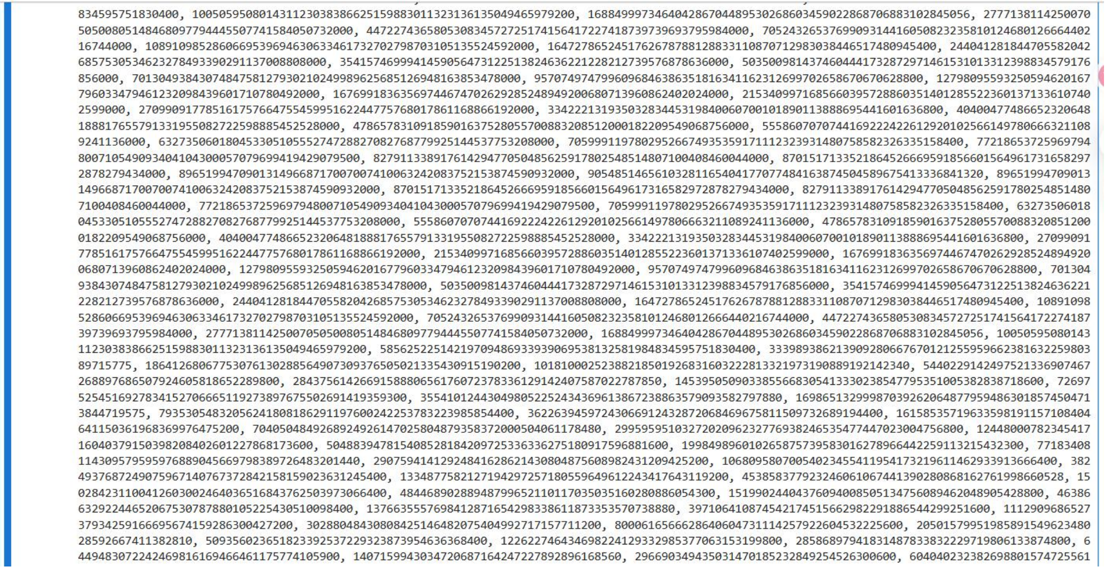

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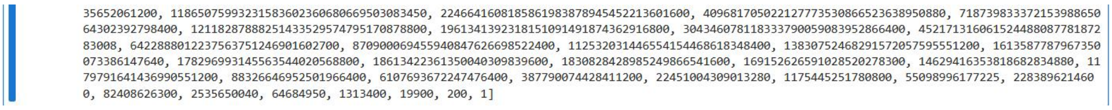

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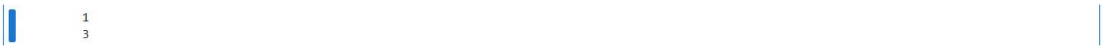
```

第四题：

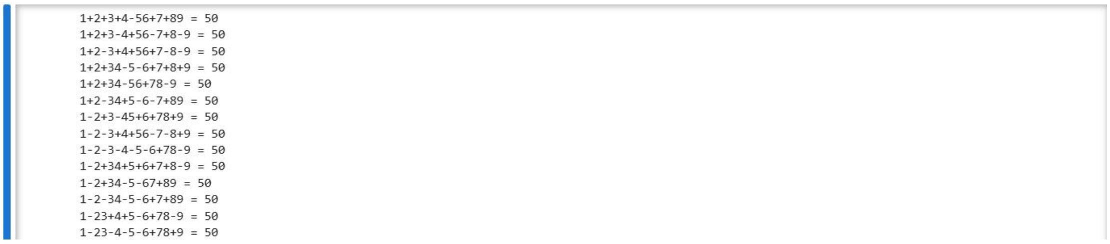
运行结果：



```
1
3
```

第五题：

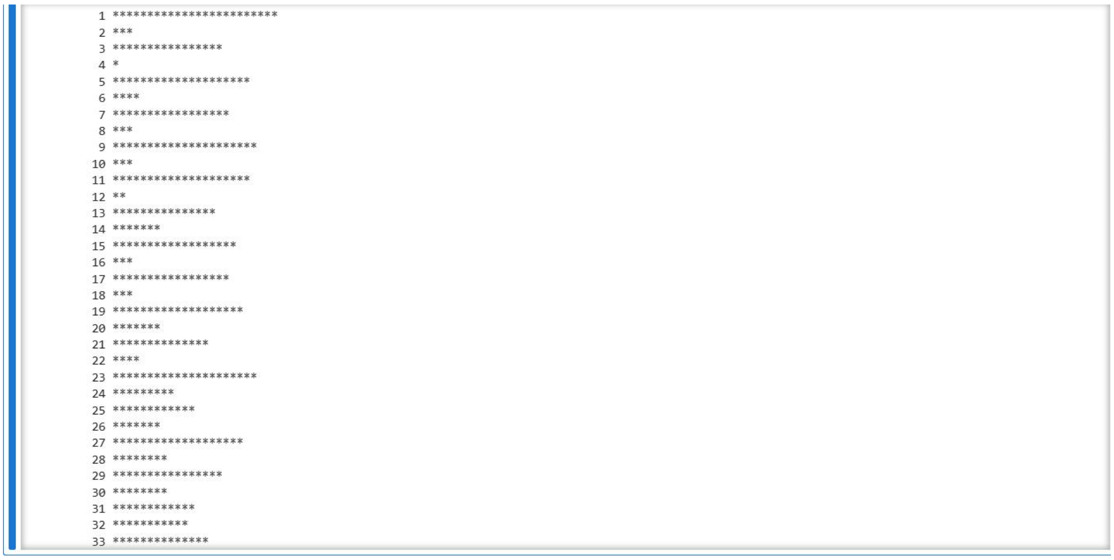
5.1 运行结果：



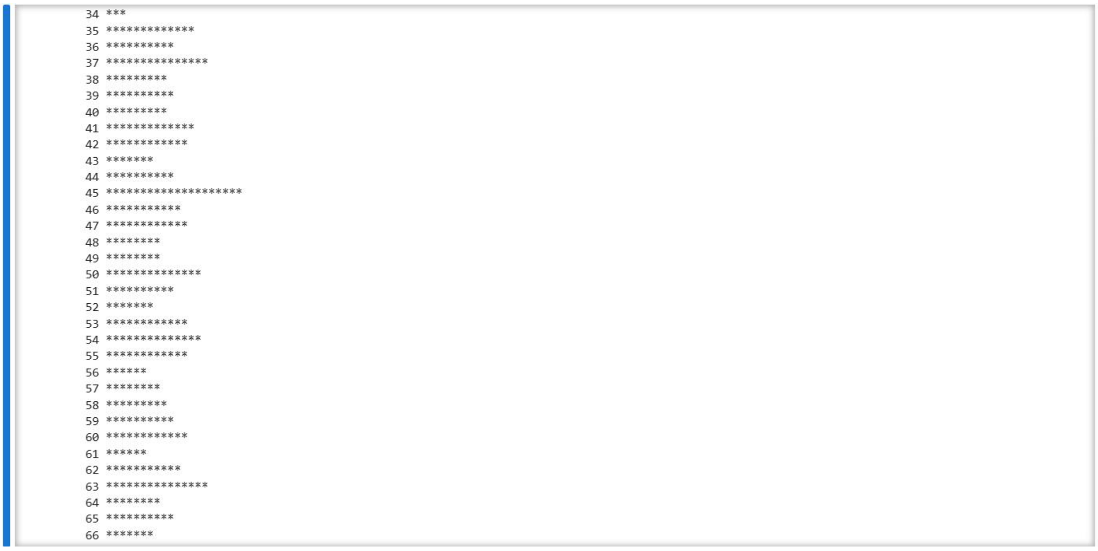
```
1+2+3+4-56+7+89 = 50
1+2+3-4+56-7+8-9 = 50
1+2-3+4+56+7-8-9 = 50
1+2+34-5-6+7+8+9 = 50
1+2+34-56+78-9 = 50
1+2-34+5-6-7+89 = 50
1-2+3-45+6+78+9 = 50
1-2-3+4+56-7-8+9 = 50
1-2-3-4-5-6+78-9 = 50
1-2+34+5+6+7+8-9 = 50
1-2+34-5-67+89 = 50
1-2-34-5-6+7+89 = 50
1-23+4+5-6+78-9 = 50
1-23-4-5-6+78+9 = 50
```

我是通过观看[给表达式添加运算符动画图解 LeetCode282 | 吴师兄图解算法 · AlgoMooc](#)获得了灵感，了解递归函数 dfs 的逻辑计算。

5.2 运行结果：



```
 1 ************************
 2 ***
 3 ****************
 4 *
 5 ********************
 6 ****
 7 *****************
 8 ***
 9 *********************
10 ***
11 ********************
12 **
13 ***************
14 *******
15 ******************
16 ***
17 *****************
18 ***
19 *******************
20 *******
21 **************
22 ****
23 *********************
24 *********
25 ************
26 *******
27 *******************
28 ********
29 ****************
30 ********
31 ************
32 ***********
33 **************
```



```
 34 ***
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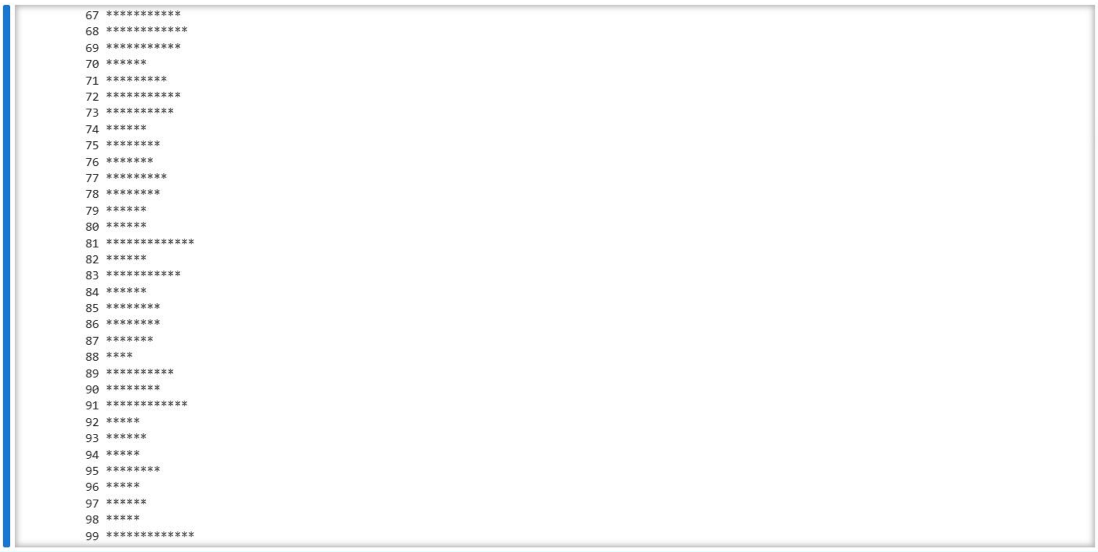
 36 **********
 37 ***************
 38 *********
 39 **********
 40 *********
 41 *************
 42 ************
 43 *******
 44 **********
 45 ********************
 46 ***********
 47 ************
 48 ********
 49 ********
 50 **************
 51 **********
 52 *******
 53 ************
 54 **************
 55 ************
 56 ******
 57 ********
 58 *********
 59 **********
 60 ************
 61 ******
 62 ***********
 63 ***************
 64 ********
 65 **********
 66 *******
```



```
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 70 ******
 71 *********
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 73 **********
 74 ******
 75 ********
 76 *******
 77 *********
 78 ********
 79 ******
 80 ******
 81 *************
 82 ******
 83 ***********
 84 ******
 85 ********
 86 ********
 87 *******
 88 ****
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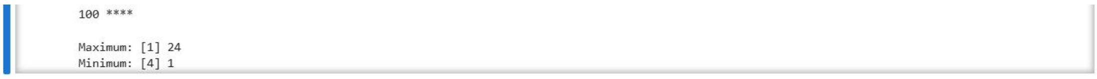

```
100 ****

Maximum: [1] 24
Minimum: [4] 1
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